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SUPERSEDING
ARMY 96-111-64
2 AUGUST 1945

MILITARY SPECIFICATION

DYE, GREEN (FOR COLORED SMOKE MIXTURES)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for the use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers dye for use in green smoke mixtures.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366 —Sieves; Standard, Testing.

MILITARY

JAN-P-122 —Packaging and Packing for Overseas Shipment — Barrels, Slack.

MIL-D-3277 —Dye, 1, 4-Di-p-Toluidinoanthraquinone (For Green Smoke Mixtures).

MIL-A-3664 —Auramine Hydrochloride (For Colored Smoke Mixtures).

U. S. ARMY

96-111-92 —Grenade, Smoke, Colored, M18.

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129—Marking of Shipments.
MIL-STD-525—Green Smoke—IV.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

INTERSTATE COMMERCE COMMISSION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Composition.—The dye shall be a uniform mixture of auramine hydrochloride conforming to Specification MIL-A-3664 and

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GPO—O—Army 60

1, 4–di–p–toluidinoanthraquinone conforming to Specification MIL–D–3277, combined in such proportions that the amount of auramine hydrochloride in the final mixture shall be not less than 27.3 nor more than 31.3 percent when tested as specified in 4.4.2.2 and the amount of 1, 4–di–p–toluidinoanthraquinone in the final mixture shall be not less than 68.7 nor more than 72.7 percent when tested as specified in 4.4.2.1. (The manufacturer shall furnish a ½-pound sample of the 1, 4–di–p–toluidinoanthraquinone and the auramine hydrochloride used in the preparation of each lot of green dye for use in determining the purity of the components.)

3.2 Volatile matter.—The volatile matter of the material shall be not more than 2.5 percent when tested as specified in 4.4.3.

3.3 Particle size.—The particle size of the material shall be such that not less than 99 percent will pass through a No. 60 U.S. Standard sieve, not less than 95 percent will pass through a No. 100 U.S. Standard sieve, and not less than 40 percent will pass through a No. 200 U.S. Standard sieve when tested as specified in 4.4.4.

3.4 Apparent density. — The apparent density shall be 0.35 ± 0.15 when tested as specified in 4.4.5.

3.5 Performance. — The material shall burn without flaming and shall emit a continuous cloud of green smoke for a period of 70 ± 20 seconds when tested as specified in 4.4.6. The smoke shall be distinguishable as to color, against a contrasting color background, at a distance of 10,000 feet on a clear day (see 6.3).

3.6 Workmanship.—All work shall be done in a neat and workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.—A lot shall consist of material produced by one manufacturer with no change in process or materials in not more than 24 consecutive hours, providing the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.4).

4.2 Sampling.—From each of five containers taken at random from each lot, two representative specimens shall be removed as follows:

Specimen A: For physical and chemical tests—1 pound (see 4.4.2 through 4.4.5).

Specimen B: For performance test—2 pounds (see 4.4.6).

Each specimen taken shall be placed in a clean, dry container, sealed, and labeled in such a manner as to identify the specimen with the prime container and the lot represented (see 6.2).

4.3 Inspection.

4.3.1 *Components.* — The inspector shall ascertain that prior to mixing, the components of the dye mixture have been inspected, tested, and accepted in accordance with their respective specifications.

4.3.2 *Packing and marking.*—The inspector shall inspect the packing and marking of the material for compliance with section 5.

4.3.3 *Procedures.*—For Naval purchases, general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 Tests.

4.4.1 *Reagents.*—Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and corrections shall be applied when significant.

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4.4.2 *Composition.*

4.4.2.1 *Dye, 1,4-di-p-toluidinoanthraquinone.*—Weigh, to the nearest milligram (mg.), 0.361 gram (gm.) of the specimen that has been previously dried to constant weight in an oven maintained at 70 to 75°C. and transfer quantitatively to a 500-milliliter (ml.) beaker, using small amounts of benzene for washing. Dissolve in approximately 200 ml. of benzene. Stir and break up all lumps to insure complete wetting and to facilitate solution. Transfer quantitatively to a 500-ml. volumetric flask, dilute to the mark with benzene, and mix thoroughly. Filter through a 15-centimeter (cm.) fluted filter paper and discard the first 300 ml. of the filtrate. Remove a 20-ml. aliquot from the remaining 200 ml. of the filtrate, transfer quantitatively to a second 500-ml. volumetric flask, dilute to the mark with benzene and mix thoroughly. Transfer a portion of the solution to the spectrophotometer cell and fill a reference cell with benzene from the same source as that used in the preparation of the solution. Read the transmittance (T) of the solution at a wave length of 642 millimicrons.

$$\text{Percent 1, 4-di-p-toluidinoanthraquinone} = \frac{-100 \log_{10} T}{P}$$

where:

P=purity of the 1, 4-di-p-toluidinoanthraquinone used in the preparation of the dye.

Note.—The above procedure and formula apply when: the spectrophotometer cell depth is 10 millimeter (m.m.), the transmittance (T) is expressed as a fraction less than 1, and the spectrophotometer is a Hardy-General Electric recording spectrophotometer having an effective band width of 10 millimicrons. When another instrument is used, it shall first be calibrated against a Hardy-General Electric instrument or against U. S. Bureau of Standards glass filters. When a spectrophotometer cell of different depth is used, calculate the weight of the specimen in inverse proportion.

4.4.2.2 *Auramine hydrochloride.*—Weigh, to the nearest mg., 0.346 gm. of the specimen that has been previously dried to constant weight in an oven maintained at 70 to 75°C. and transfer quantitatively to a 600-ml. beaker. Dissolve in approximately 400 ml. of water (not warmer than 30°C.). Stir to insure complete wetting and to facilitate complete solution of the water-soluble constituent. Do not use a wetting agent. Transfer quantitatively to a 1000-ml. volumetric flask. Fill the flask to the mark with water and mix thoroughly, filter through a 15-cm. fluted filter paper and discard the first 600 ml. of the filtrate. Remove a 10-ml. aliquot from the remaining 400 ml. of the filtrate and transfer to a 250-ml. volumetric flask. Make up to the mark with water, and mix thoroughly. Within 1 hour after preparation of the solution, transfer a portion of the solution to a spectrophotometer cell and fill a reference cell with water from the same source as that used in the preparation of the solution. Read the transmittance (T) of the solution at a wave length of 430 millimicrons.

$$\text{Percent auramine hydrochloride} = \frac{-100 \log_{10} T}{2P}$$

where:

P=purity of the auramine hydrochloride used in the preparation of the dye.

Note.—The above procedure and formula apply when: the spectrophotometer cell depth is 10 mm., the transmittance (T) is expressed as a fraction less than 1, and the spectrophotometer is a Hardy-General Electric recording spectrophotometer having an effective band width of 10 millimicrons. When another instrument is used, it shall first be calibrated against a Hardy-General Electric instrument or against U. S. Bureau of Standards glass filters. When a spectrophotometer cell of different depth is used, calculate the weight of the specimen in inverse proportion.

4.4.3 *Volatile matter.*—Weigh, to the nearest milligram (mg.), approximately 5 gm. of the specimen in a tared, glass-stoppered weighing bottle approximately 50 mm. in diameter by 30 mm. high. Dry to constant weight in an oven maintained at 70 to 75°C., cool in a desiccator, and weigh.

$$\text{Percent volatile matter} = \frac{\text{Loss in weight} \times 100}{\text{weight of specimen}}$$

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4.4.4 *Particle size.*—Numbers 60, 100, and 200 U.S. Standard sieves, conforming to Specification RR–S–366, shall be selected for the test. Nest the sieves in order of decreasing mesh size, placing the largest mesh on top and a pan at the bottom. Weigh, to the nearest centigram (cg.), approximately 10 gm. of the material and transfer to the No. 60 sieve. Brush the material with a camel's-hair brush until no more material passes through the sieve. Remove the sieve and weigh the material left on the sieve. Repeat the procedure with the material on the No. 100 and No. 200 sieves. Calculate the percent material passing through each sieve by subtracting from 100 percent the sum of the percentages of the material retained on that sieve and all of the larger mesh sieves used.

4.4.5 *Apparent density.*

4.4.5.1 *Apparatus.*—The apparatus shall consist of a cylinder and a glass funnel. The cylinder shall have an internal length of 11 ± 1 cm. and a volume of 100 ml. and may be made by cutting off a 250-ml. graduate at the 100-ml. graduation. The glass funnel shall be approximately 9 cm. in diameter at the top and shall have a stem approximately 2.5 cm. in length and 1.5 cm. in inside diameter.

4.4.5.2 *Procedure.*—Weigh the cylinder. Support the funnel so that the end of the stem is centrally located 1 cm. above the top of the cylinder. Dry, to a constant weight in an oven maintained at 70 to 75°C., a portion of the specimen sufficient to fill the cylinder to overflowing. Shake the specimen from a paper into the funnel at a rate of approximately 50 ml. per minute. Avoid jarring the cylinder during the filling operation. If clogging occurs in the funnel, return the contents to the paper and repeat the filling until the material overflows the cylinder. Strike off the material level with the top of the cylinder and weigh the cylinder and the contents. Calculate the apparent density of the specimen by dividing the increase in weight of the cylinder by 100.

4.4.6 *Performance.*—Prepare and fire six green smoke grenades in accordance with Specification 96–111–92, using the smoke mixture specified in Standard MIL–STD–525. Upon firing, the grenades thus prepared shall conform to 3.5 (see 6.3 and 6.5).

4.5 **Rejection and resubmission.** — If any specimen, selected as specified in 4.2, fails to conform to this specification, the lot shall be rejected. The contractor, at no expense to the Government, shall have the option of having a partial or complete analysis made of each container in the lot and removing the nonconforming material. The contractor may then resubmit the remaining portion of the lot for testing.

5. PREPARATION FOR DELIVERY

5.1 **Packing.**—The material shall be packed either in steel drums conforming to Specification 37D of Regulations for Transportation of Explosives and Other Dangerous Articles, etc., in fiber drums conforming to Specification 21A or 21B of Regulations for Transportation of Explosives and Other Dangerous Articles, etc., for an authorized 200-pound net weight, or in No. 1 wood barrels conforming to Specification JAN–P–122. Each drum or barrel shall be fitted with a polyethylene bag constructed of polyethylene sheet having a nominal thickness of 0.004 inch. After filling, the bag shall be closed by knotting, tying, or heat sealing.

5.2 **Marking.**—Each container shall be conspicuously labeled:

Caution:—Avoid skin contact or breathing of dust or fumes. Provide adequate ventilation in work areas.

In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 **Ordering data.** — Procurement documents should specify the title, number, and date of this specification.

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6.2 Sampling and testing.—When the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the contractor, to the sampling and testing procedures specified in this specification.

6.3 Alternate requirements.—Alternate performance requirements and tests, simulating the end use of the material, may be substituted for those specified in 3.5 and 4.4.6. However, in case of dispute, the requirements and test specified in 3.5 and 4.4.6 will govern.

6.4 Batch.—A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.5 Evaluation samples.—Because of the special nature of the equipment required in preparing colored smoke grenades, the contractor may submit to the procuring agency, samples of the material he proposes to furnish for evaluation as to performance. However, the results of such tests will be considered informative only and in no sense will they be construed as acceptance tests.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Chemical Corps

Other interest:
Army—O
Navy—Or.